

Building a Coherent Proof from the Givens

Justifying each step of an algebraic solution, building an equation from a segment argument, building one from an angle argument, and writing a complete two-column proof

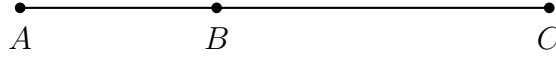
Directions.

- Every line you write needs a reason. “Given”, a definition, a postulate, or a property of equality — one of those, named.
- Start from what is *stated*. A diagram can suggest a relationship, but only the givens and what follows from them may be used as reasons.
- When a problem asks for a value, show the equation you built *and* say which definition or postulate licensed it.
- Angle measures are in degrees.

1. Solve $3x + 7 = 22$, writing one equation per line. Beside each line, name the property of equality that justifies it.

Answer: _____

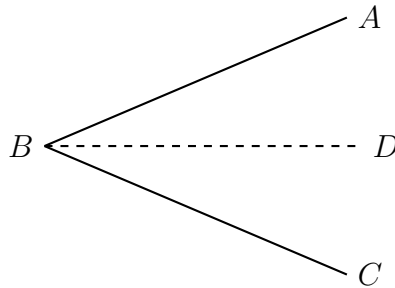
2. Point B lies between A and C on $\overline{AC}$, with $AB = 2x + 3$, $BC = 3x - 1$, and $AC = 27$.



Name the postulate that lets you combine the three givens into one equation. Then find x , and use it to find AB .

Answer: _____

3. $\overrightarrow{BD}$ bisects $\angle ABC$. The two halves measure $m\angle ABD = (2x + 5)^\circ$ and $m\angle DBC = (4x - 19)^\circ$.



State the definition that makes the two expressions equal, find x , and then find $m\angle ABC$.

Answer: _____

4. Solve $5(x - 2) = 3x + 8$. Write one equation per line, and name the property that justifies each line. One of your reasons is *not* a property of equality — say which line it is on and what it is called instead.

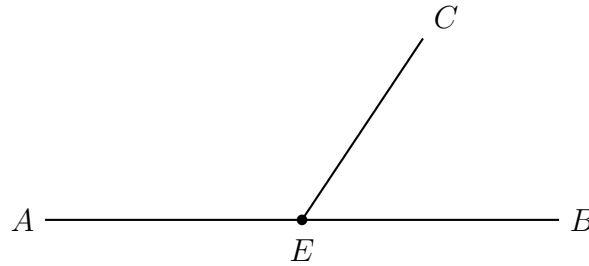
Answer: _____

5. **Given:** M is the midpoint of $\overline{AB}$, and N is the midpoint of $\overline{AM}$. **Prove:** $AN = \frac{1}{4}AB$.



Statements	Reasons

6. $\angle AEC$ and $\angle CEB$ form a linear pair, with $m\angle AEC = (3x + 10)^\circ$ and $m\angle CEB = (x + 30)^\circ$.



Name the postulate that tells you what the two measures add to, then find x and $m\angle AEC$.

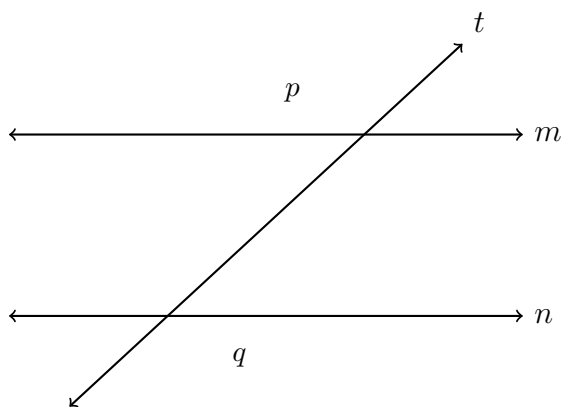
Answer: _____

7. Point P lies between Q and R , with $QP = 4x - 5$, $PR = 2x + 7$, and $QR = 50$. Dana started her solution by writing $4x - 5 = 2x + 7$.

- (a) Explain, in one or two sentences, why that equation does not follow from the givens.
(b) Write the equation that *does* follow, name the postulate behind it, and solve for x .

Answer: _____

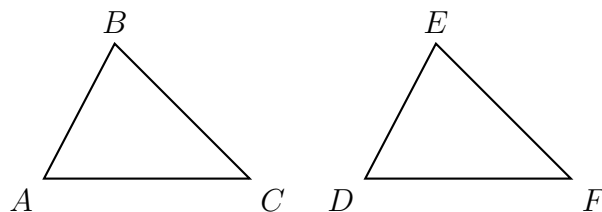
8. Lines m and n are parallel and are cut by transversal t . The two marked angles, $\angle p$ and $\angle q$, are same-side interior angles, with $m\angle p = (5x - 12)^\circ$ and $m\angle q = (3x + 16)^\circ$.



Name the theorem that relates $\angle p$ and $\angle q$, then find x , $m\angle p$, and $m\angle q$.

Answer: _____

9. **Given:** $\overline{AB} \cong \overline{DE}$, $\overline{BC} \cong \overline{EF}$, and $\angle B \cong \angle E$. **Prove:** $\overline{AC} \cong \overline{DF}$.



Write a two-column proof. Your last reason should explain how corresponding parts follow once the triangles are congruent.

Statements	Reasons

10. Synthesis. $\overrightarrow{BD}$ is drawn inside $\angle ABC$ so that $\angle ABD \cong \angle DBC$. It is also given that $m\angle ABC = (7x - 4)^\circ$ and $m\angle ABD = (2x + 10)^\circ$.

(a) Write a short argument, in complete sentences, that starts from the given congruence and ends with the equation $7x - 4 = 2(2x + 10)$. Name every definition or postulate you use, in the order you use it.

(b) Solve your equation for x , then find $m\angle ABC$.

Answer: _____